21-242 Recitation 12

RREF — Q7. RREF when T A is injective

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

If T_A is injective, what does $\mathrm{RREF}(A)$ look like?

Course-level solution

$T_A : \mathbb{F}^n \to \mathbb{F}^m$ injective means $\ker T_A = \{\vec{0}\}$, so there are no free variables: every column is a pivot column. That requires $n \leq m$, and

$$\mathrm{RREF}(A) = \begin{pmatrix} I_n \\ 0_{(m-n)\times n} \end{pmatrix}$$

(the block 0 is absent if $m = n$, in which case $\mathrm{RREF}(A) = I_n$).